

Exam Date & Time: 11-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech IV Semester - End Semester Examination - April / May -2024

Machine Learning in VLSI computer aided design [ECE-4310]

Marks: 50

Duration: 180 mins.

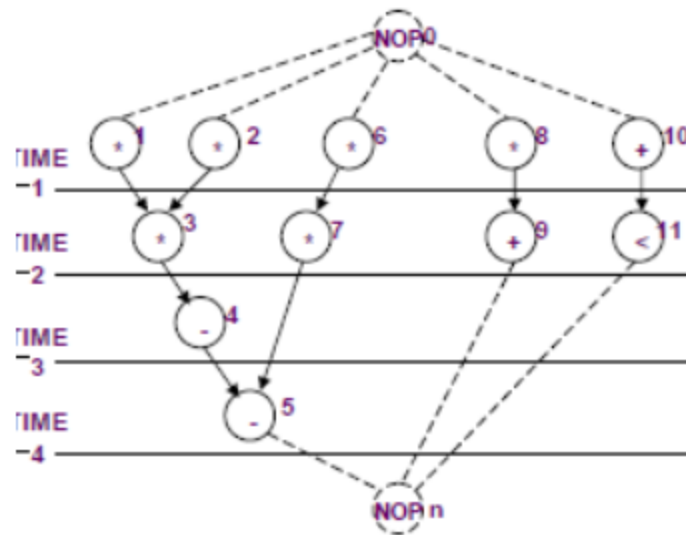
PART-A

Answer all the questions.

- 1) Implement Transparent high latch and low latch using ACT-2S logic module (4)
 - A)
 - B) Implement $F=A'B+ABC'+A'B'C$ using ACT-1 logic module (3)
 - C) Explain VLSI design flow (3)
- 2) Draw ROBDD for 3-bit synchronous even counter. Show all the steps starting from OBDD. Perform ITE algorithm for ROBDD. (4)
 - A)
 - B) Draw ROBDD for 2 input XOR gate starting from BDD. Show all the steps. (3)
 - C) Draw Control and data flow graph for the following sequential statements $X=A*B$, $Y=C+D$, if $X < Y$ then $Z=Y-X$ else $Z=X-Y$. (3)
- 3) Find the essential prime implicant for the set of prime implicants $F=\{C_1, C_2, C_3, C_4\}$ where $C_1=x_1'x_3'x_4'$, $C_2=x_1x_2$, $C_3=x_3x_4$, $C_4=x_2x_3'x_4'$ using ESPRESSO algorithm. (4)
 - A)
 - B) Determine the prime implicant for the following function using iterated consensus method. (3)

$$F(A,B,C) = A'B'C'+A'BC'+A'BC+AB'C'+AB'C+ABC'$$
 - C) Define the term exact minimization and heuristic minimization with an example each. (3)
- 4) (4)
 - A)

Apply LIST-L Scheduling algorithm for the given data flow graph shown in Figure. Assume $\gamma=4$, $M_1=M_2=M_3=M_4=M_5=0$; $M_6=M_7=1$; $M_8=M_9=M_{10}=M_{11}=2$. Draw the scheduled graph under resource constraints.



- B) Explain ALAP algorithm with an example (3)
- C) Define latency, mobility and resource constraints (3)
- 5) Implement the following function $F=A'CD+B'CD+A'B+BC'$ using PAL (4)
- A)
- B) Draw the Sequencing graph for the function $F=A'B+AB'$ (3)
- C) Explain about Data Flow Graph (DFG) and Sequencing graph (3)

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